SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Software Requirement Specification (SRS)

Scenario-Based Requirement Analysis — CO2 Assessment Tool

Intelligent AI-Based Airport Baggage Tracking and Management System

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This document presents a scenario-based requirement analysis and Software Requirement Specification for an AI-powered airport baggage tracking and management platform. It covers the problem context, functional and non-functional requirements, actor and use-case modeling, data and interface requirements, system features, assumptions, constraints, and validation of requirement completeness and consistency.

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements for the proposed **Intelligent AI-Based Airport Baggage Tracking and Management System**. The purpose of the system is to provide airports, airlines, ground-handling staff, and passengers with a unified platform for real-time baggage visibility, predictive delay management, and operational analytics, thereby reducing baggage mishandling and improving passenger satisfaction.

1.2 Business Domain and Context

Airports handle millions of passenger bags annually, and even a small percentage of mishandled bags translates into significant financial loss and reputational damage for airlines and airport operators. The domain combines airport ground operations, airline IT systems, IoT/RFID sensing infrastructure, and data analytics. Modern baggage systems are expected to integrate RFID scanning, IoT-enabled conveyor and sorting infrastructure, cloud-based data platforms, and AI/ML models to move from reactive baggage handling toward proactive, predictive logistics management.

1.3 Stakeholders

Stakeholder	Interest / Role
Passengers	Want accurate, real-time visibility of their baggage status and timely notifications about delays or misrouting.
Airport Ground Staff / Baggage Handlers	Operate scanning, sorting, and loading processes; need accurate tracking and routing information to act quickly.
Airlines	Responsible for baggage service quality, compensation costs, and coordination with airport operators.
Airport Operations / IT Administrators	Manage system configuration, users, integrations with airport infrastructure, and monitor operational KPIs.
Regulatory Bodies	Require compliance with aviation safety, security, and data-protection regulations.

1.4 System Objectives

- Track baggage location and status in real time from check-in to final delivery.
- Optimize baggage routing across sorting and transfer points to reduce mishandling.
- Apply AI/ML models to predict potential baggage delays before they occur.
- Automatically notify passengers of baggage status changes and delay events.
- Reduce baggage misrouting and improve first-time-right delivery rates.
- Generate operational analytics and reports for airport and airline management.
- Improve overall airport logistics efficiency and passenger satisfaction.

1.5 Existing Challenges

- Traditional barcode-based systems provide limited real-time tracking granularity.
- Baggage status information is fragmented across airline, airport, and ground-handler systems.
- Delay and misrouting risks are typically identified only after they occur, not predicted in advance.
- Passengers have little visibility into baggage status, leading to complaints and support load.
- Manual reconciliation of baggage records increases operational cost and error rate.

2. Problem Description

Baggage delays, loss, and misrouting create significant inconvenience for passengers and increase operational costs for airlines. Existing baggage handling systems offer limited real-time tracking and little to no predictive capability, so problems are usually detected after a bag has already been delayed or misrouted. The proposed system addresses this gap by combining RFID/IoT-based real-time tracking, AI-driven delay prediction, automated passenger notification, and operational analytics into a single platform, with the goal of improving baggage handling efficiency while reducing passenger complaints and operational cost.

3. Scope of the System

The system scope covers baggage check-in/registration, RFID/IoT-based real-time tracking across airport zones (check-in, sorting, transfer, loading, arrival), AI-based delay prediction and route optimization, automated passenger notification (SMS/app/email), a staff-facing operations dashboard, and an administrative reporting and analytics module. The system integrates with airline and airport operational systems but does not replace core airline reservation, ticketing, or physical conveyor/sorter hardware control systems — it consumes and acts upon data from these systems through defined interfaces.

In scope:

- Baggage registration, RFID tagging reference, and check-in linkage to passenger/flight data.
- Real-time baggage location tracking across airport processing zones.
- AI/ML-based prediction of baggage delay risk and route optimization suggestions.
- Automated passenger notifications for status changes, delays, and misrouting alerts.
- Staff dashboard for monitoring baggage flow and manually rerouting flagged baggage.
- Administrative reporting, analytics, and system/user management.

Out of scope:

- Physical control of conveyor belts, sorters, or robotic handling hardware.
- Airline ticketing, check-in kiosk hardware, and fare/reservation management.
- Payment processing for excess baggage or compensation disbursement.

4. Functional Requirements

ID	Functional Requirement
FR-1	The system shall allow airline/ground staff to register baggage and link it to a passenger and flight record at check-in.
FR-2	The system shall capture baggage scan events from RFID/IoT readers at each processing checkpoint in real time.
FR-3	The system shall display the current location and status of any registered baggage to authorized users.
FR-4	The system shall use an AI/ML model to predict the probability of baggage delay based on historical and live operational data.
FR-5	The system shall recommend optimized baggage routing paths across sorting and transfer points.
FR-6	The system shall automatically send notifications to passengers on baggage check-in, loading, delay, and arrival events.
FR-7	The system shall allow staff to flag and manually reroute baggage identified as misrouted or at risk.

ID	Functional Requirement
FR-8	The system shall allow passengers to raise and track lost/delayed baggage complaints.
FR-9	The system shall generate operational reports on baggage volume, delay rates, and misrouting incidents.
FR-10	The system shall support authentication and role-based access control for passengers, staff, and administrators.
FR-11	The system shall allow administrators to manage users, roles, and system configuration.
FR-12	The system shall log all baggage events for audit and analytics purposes.

5. Non-Functional Requirements

Category	Requirement
Performance	The system shall process and reflect an RFID scan event in the tracking dashboard within 5 seconds under normal load.
Scalability	The system shall support concurrent tracking of at least 500,000 baggage items per day across multiple terminals without degradation.
Reliability	The system shall maintain 99.5% uptime for core tracking and notification services during airport operating hours.
Security	The system shall encrypt passenger and baggage data in transit and at rest, and enforce role-based access control.
Usability	The staff dashboard and passenger app shall be usable with no more than 30 minutes of training for new staff.
Maintainability	The system shall use modular, service-oriented components to allow independent updates to the AI model and notification service.
Availability	Core tracking and notification services shall be available 24/7, including during peak travel periods.
Interoperability	The system shall expose standard APIs to integrate with airline and airport operational systems.
Data Retention	Baggage event and tracking data shall be retained for a minimum of 12 months to support analytics and audits.

6. Actors and Use Cases

6.1 Actors

Actor	Type	Description
Passenger	Primary	Checks baggage status, receives notifications, raises complaints.
Airport Staff / Baggage Handler	Primary	Scans baggage, monitors sorting, manually reroutes flagged bags.
System Administrator	Primary	Manages users, roles, and system configuration; views reports.
RFID / IoT Sensors	Secondary	Automated hardware actors that generate baggage scan events.
Airline Ops System	Secondary	External system providing flight and passenger data via API.
AI Prediction Engine	Secondary	Internal/external ML service that computes delay predictions.

6.2 Use Case Diagram

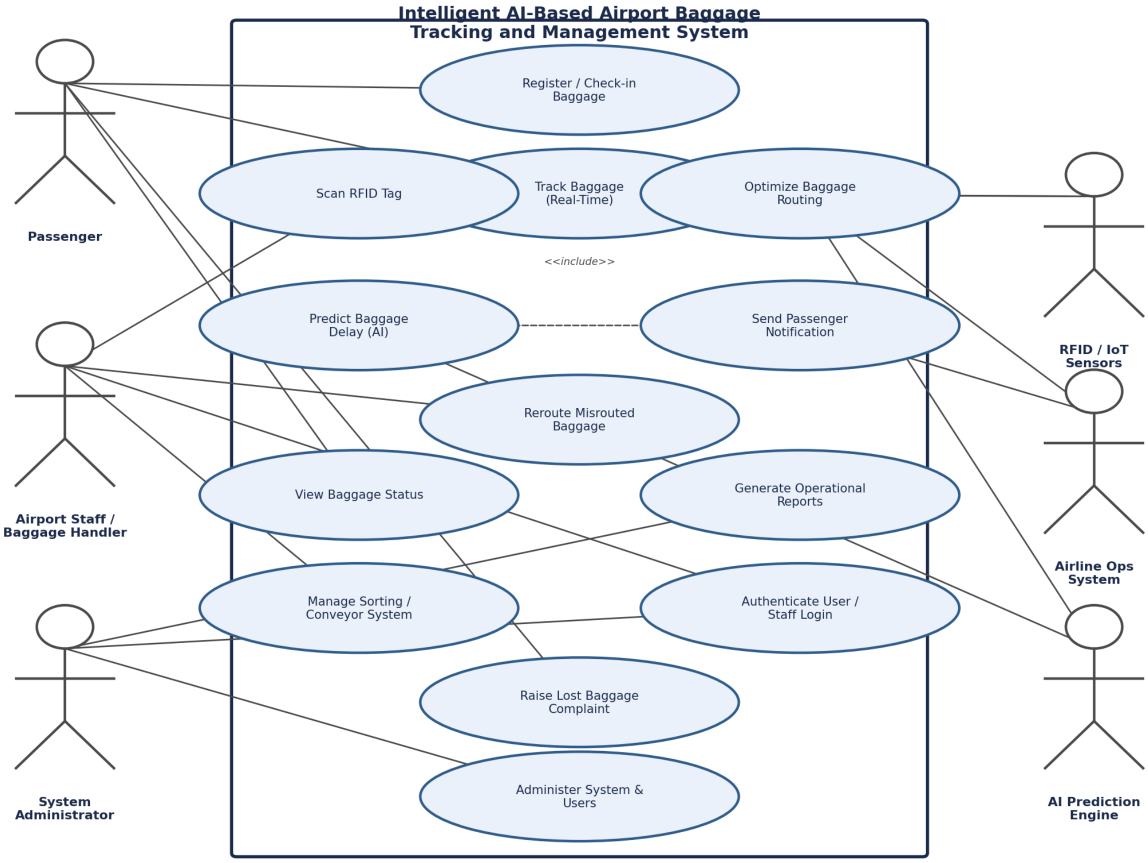

Figure 1: Use Case Diagram — Intelligent AI-Based Airport Baggage Tracking and Management System

6.3 Use Case Descriptions

Use Case	UC-1: Track Baggage in Real Time
Actors	Passenger, Airport Staff, RFID/IoT Sensors
Preconditions	Baggage has been registered and tagged at check-in.
Main Flow	1. RFID/IoT sensors scan the baggage tag at a checkpoint. 2. The system records the scan event with timestamp and location. 3. The tracking engine updates the baggage status. 4. Passenger or staff views the updated status on the app/dashboard.
Postcondition	Baggage status and location are updated and visible to authorized users.

Use Case	UC-2: Predict Baggage Delay
Actors	AI Prediction Engine, Airport Staff
Preconditions	Baggage tracking data and flight transfer information are available.
Main Flow	1. The AI engine analyzes live tracking data, transfer time windows, and historical patterns. 2. The engine computes a delay-risk score for each in-transit bag. 3. If risk exceeds a threshold, the system raises an alert to staff and triggers a notification workflow.
Postcondition	At-risk baggage is flagged before the delay occurs, enabling proactive handling.

Use Case	UC-3: Notify Passenger of Baggage Status
Actors	Passenger, Notification Service
Preconditions	Passenger has an active baggage record linked to a valid contact channel.
Main Flow	1. A tracked baggage event (check-in, loaded, delayed, arrived) occurs. 2. The notification service formats a status message. 3. The message is sent via app push, SMS, or email. 4. Passenger receives and can view detailed status in the app.
Postcondition	Passenger is informed of baggage status changes without manual inquiry.

7. Data Requirements

The system must capture, process, and persist several categories of data to support real-time tracking, prediction, and reporting. Key data entities include Passenger, Flight, Baggage, Tracking Event, Prediction Result, and Notification, related as shown conceptually in the data flow diagram below.

Entity	Key Attributes
Passenger	Passenger ID, name, contact details, flight reference.
Flight	Flight number, origin, destination, scheduled/actual times, transfer windows.
Baggage	Baggage tag ID (RFID), passenger reference, flight reference, weight, current status.
Tracking Event	Event ID, baggage tag ID, checkpoint/location, timestamp, sensor ID.
Prediction Result	Baggage tag ID, delay-risk score, contributing factors, generated timestamp.
Notification	Notification ID, passenger ID, message type, channel, delivery timestamp/status.

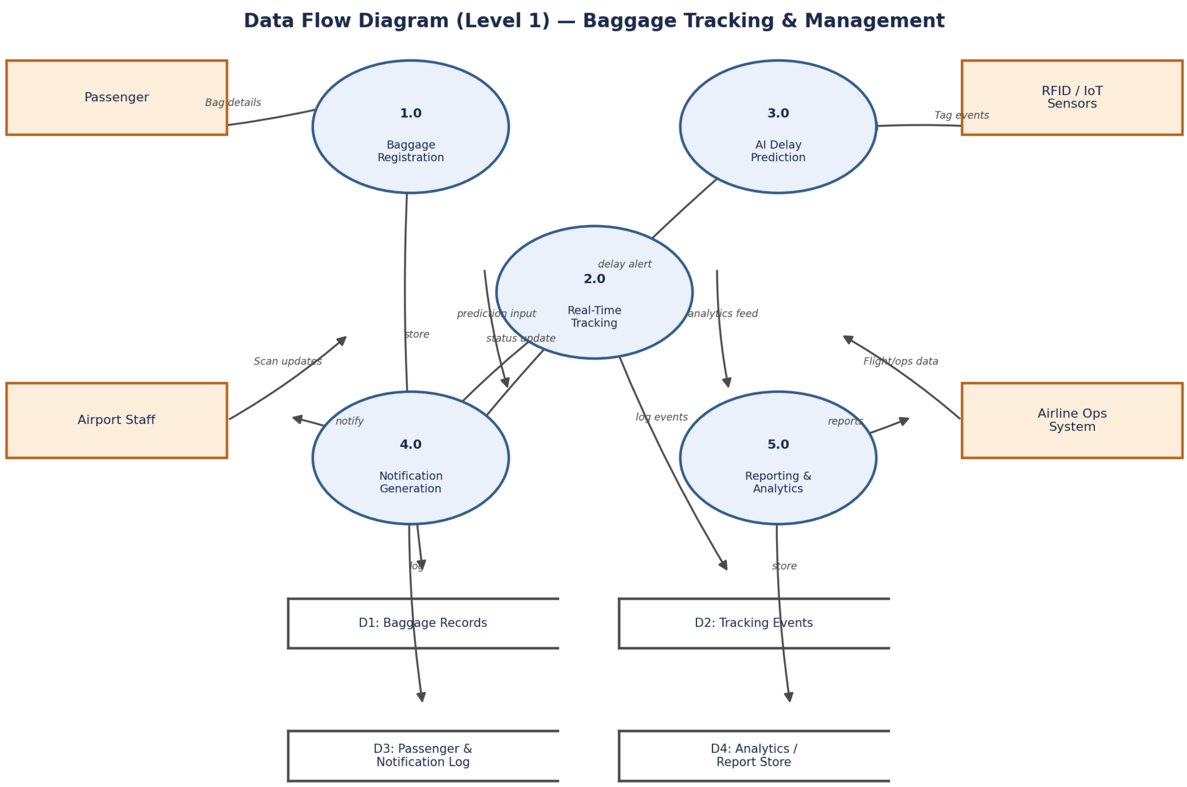

Figure 2: Level-1 Data Flow Diagram — Baggage Tracking and Management

8. External Interface Requirements

8.1 User Interfaces

- Passenger mobile/web app for baggage status viewing, notifications, and complaint filing.
- Staff/ground-operations dashboard for real-time monitoring and manual rerouting.
- Administrator console for user management, configuration, and report generation.

8.2 Hardware Interfaces

- RFID readers and IoT gateways installed at check-in, sorting, transfer, and loading points.
- Conveyor/sorting-system sensors providing baggage movement and jam/exception signals.

8.3 Software Interfaces

- REST/API integration with airline reservation and departure-control systems for flight and passenger data.
- Messaging/notification gateway (SMS, email, push notification) for passenger alerts.
- AI/ML model-serving interface for delay prediction and route optimization scoring.

8.4 Communication Interfaces

The system shall communicate over secure HTTPS/TLS channels for all API and application traffic, and use a message queue (e.g., MQTT/Kafka-style protocol) for high-frequency IoT sensor event ingestion.

9. System Features

The system is organized as a layered architecture: sensor/data-capture layer, ingestion and messaging layer, AI/processing layer, core application/business-logic layer, data-storage layer, and presentation layer for passengers, staff, and administrators, as shown below.

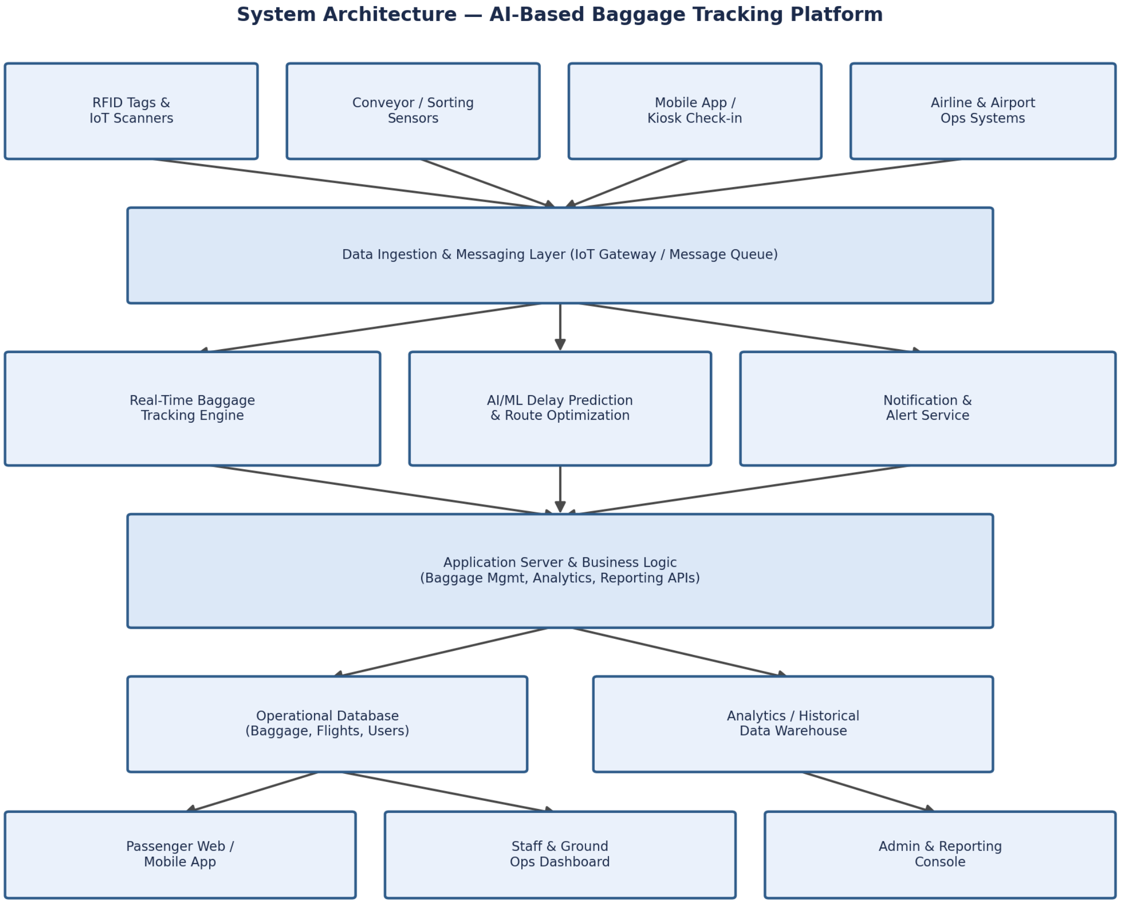

Figure 3: System Architecture — AI-Based Baggage Tracking Platform

Feature	Description
Real-Time Tracking Engine	Consumes RFID/IoT scan events and maintains a live baggage-location state.
AI Delay Prediction & Routing	Applies ML models to historical and live data to predict delays and suggest optimal routing.
Notification Service	Delivers multi-channel passenger notifications based on tracked baggage events.
Reporting & Analytics	Aggregates operational data into dashboards and periodic reports for management.
Access Control & Administration	Manages authentication, roles, and configuration for all user categories.

10. Assumptions and Constraints

10.1 Assumptions

- All checked baggage is fitted with a functioning RFID tag at check-in.
- Airport terminals have reliable network coverage to support real-time IoT data transmission.
- Airline systems expose APIs (or equivalent data feeds) for flight and passenger information.
- Passengers provide a valid contact channel (mobile number/email/app) for notifications.
- Historical baggage-handling data is available in sufficient volume to train the AI prediction model.

10.2 Constraints

Type	Constraint
Technical	The system must integrate with existing legacy airport and airline IT infrastructure with limited modification.
Operational	The system must operate continuously (24/7) without disrupting live baggage-handling operations during deployment.
Economic	Implementation cost is bounded by the airport's IT modernization budget, limiting the scale of initial IoT sensor rollout.
Legal / Regulatory	The system must comply with aviation security regulations and data-protection laws governing passenger data.
Time	The initial platform must be delivered within the airport's planned modernization phase timeline.

11. Requirement Validation — Completeness and Consistency

11.1 Stakeholder Coverage Check

Each stakeholder group identified in Section 1.3 is addressed by at least one functional requirement: passengers (FR-3, FR-6, FR-8), airport staff (FR-2, FR-7), airlines and operations (FR-5, FR-9), and administrators (FR-10, FR-11). This confirms that all identified stakeholder concerns are represented in the requirement set.

11.2 Ambiguity, Redundancy, and Consistency Review

Check	Observation
Ambiguity	Terms such as “real time” and “delay risk” are quantified with explicit thresholds in the non-functional requirements (e.g., NFR-Performance) to remove ambiguity.
Redundancy	Requirements were cross-checked; no duplicate functional requirements were found. FR-2 (scan capture) and FR-3 (status display) are related but distinct and non-overlapping.
Inconsistency	No conflicting requirements were identified; notification requirements (FR-6) are consistent with the interface requirements defined in Section 8.
Missing Requirements	An initial gap was identified for audit logging; FR-12 was added to ensure all baggage events are logged for analytics and compliance.
Testability	Each functional requirement is stated as a single, verifiable system action, allowing direct mapping to test cases.

11.3 Suggested Improvements

- Define explicit SLA thresholds for each non-functional requirement to support formal acceptance testing.
- Add a requirement traceability matrix linking each functional requirement to its originating stakeholder need and test case.
- Introduce a data-quality validation requirement for RFID scan accuracy to reduce false tracking events.
- Periodically retrain and validate the AI prediction model against new operational data to maintain prediction accuracy.

11.4 Conclusion

The requirement analysis performed in this document addresses the scenario's stated objectives and expected outcomes. Functional and non-functional requirements, actor and use-case models, data and interface requirements, and system features have been validated for completeness and consistency, with identified gaps closed and improvement recommendations documented for future iterations of the specification.